# SPARK: TEST INFRASTRUCTURE PAVING THE WAY FOR A SUSTAINABLE EUROPEAN PRESENCE IN SPACE

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**Abstract**

Commercial growth in the European space transportation sector has created a shortage of agile ground propulsion test infrastructure. Emerging launcher startups, in-space mobility developers, and university researchers often face high costs and long lead times when accessing test infrastructure. In response, the Chair of Space Mobility and Propulsion at the Technical University of Munich (TUM) established the SPARK test facility at the research airport Oberpfaffenhofen. Following the conceptual roadmap presented at EUCASS 2025, this paper presents the current state and collaborative integration model implemented to transition SPARK into an active multi-user propulsion test centre.

European demand is fragmented across cryogenic methalox, storable hydrocarbons, hybrids, and green in-space propellants. Rather than committing speculative capital to a generalised test stand in an ecosystem of rapidly evolving propulsion architectures and shifting timelines, SPARK implemented an overlap co-development strategy: identifying functional intersections between partner requirements and the long-term facility vision, constructing permanent forward-compatible facility modules, and leaving standard expansion interfaces open. Central to this approach is an authoritative digital baseline anchored by a hierarchical Piping and Instrumentation Diagram (P&ID) in KiCad, a distributed Process Control System (PCS) based on Beckhoff TwinCAT, and an EtherCAT bridge that decouples facility conditioning from partner engine sequencers. A case study of integration with Rocket Factory Augsburg demonstrates the co-engineering of high- and low-pressure $GN_2$ utility networks, water and $CO_2$ fire suppression, cryogenic LOX dump lines, pyrophoric TEA/TEB ignition, staged emergency safing, and collaborative infrastructure co-investment. Finally, the paper reports on commissioned infrastructure and active test campaigns, including OP4/TOSCA hot fires, student sounding rocket testing, Delta Orbit mobile operations, and ground checkout and testing of TUM's EU-funded ASCENT rocket hopper test bench.

## 1 Introduction and Facility Evolution

Global space transportation reached unprecedented activity in 2025 with 324 orbital launches (+25% year-on-year (YoY)) and 4,556 satellites deployed (+58% YoY), yet Europe conducted only eight launches, indicating a capacity gap and structural pressures on European space access [1]. Furthermore, with more than 80% of the global upstream market captive to foreign institutional programmes, emerging launch initiatives, in-space logistics platforms, and constellation developers have created urgent demand for rapid European ground-testing options [2, 1]. Institutional surveys confirm an accelerating launch cadence, which implies an increased demand for ground operations, making access to agile, cost-effective test infrastructure a decisive competitive edge if developed across Europe. While major national and institutional sites, such as DLR Lampoldshausen and ArianeGroup Vernon, offer specialised, high-thrust testing capabilities, early-stage commercial ventures and university teams can face procedural, scheduling, and financial barriers in accessing these test facilities. To bridge this capacity gap, the Chair of Space Mobility and Propulsion (SMP) at the Technical University of Munich (TUM) established the SPARK test centre. Situated at the research airport Oberpfaffenhofen, Bavaria, the facility repurposes a historic 1970s test hall originally constructed for testing the Dassault/Dornier Alpha Jet. With reinforced concrete blast walls, 500 $m^2$ of operational floor space, and an integrated 30 m acoustic exhaust tunnel, the site provides a robust physical foundation for thruster and access-to-space propulsion testing [3], with a facility envelope engineered to scale up to 60 kN firings.

At the 11th EUCASS conference in 2025, SPARK was presented as a four-phase conceptual roadmap [3]. That initial plan envisioned a top-down design process: constructing complete, centralised fluid distribution systems, including Media Distribution Panels (MDPs), for gaseous media and Cryogenic Distribution Panels (CDPs) for liquid oxygen (LOX), liquid methane ($LCH_4$), and propane ($LC_3H_8$).

This paper describes the co-development approach selected to effectively design, install, and qualify test infrastructure including the media distribution network and commissioning of the 60 kN test position in collaboration with keystone industrial partners.

## 2 European Launch Dependence

To quantify the capability gap introduced above, a meta-analysis of European payloads launched into orbit between 2024 and 2026 was conducted ($N = 316$ payloads), comparing launch services by origin, as synthesised in Figure 1. Following the retirement of Ariane 5, the ramp-up schedule of Ariane 6, and the return-to-flight cadence of Vega-C, European satellite operators faced an acute domestic launch deficit. Over this three-year period, 291 out of 316 European payloads (92.1%) were launched on non-European vehicles, predominantly via commercial US launch providers, with only 25 payloads (7.9%) manifested on European launchers.

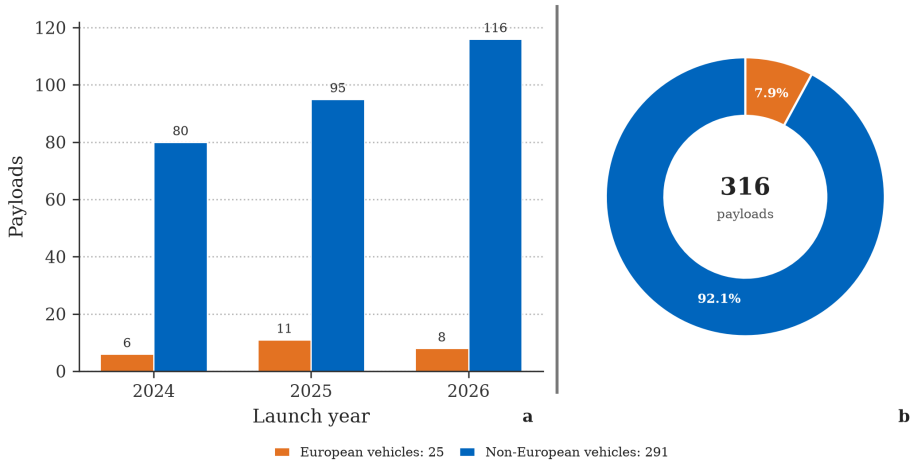



Figure 1: Meta-analysis of European satellite payload launches (2024–2026): (a) annual distribution of European payloads launched on European vs. non-European launch vehicles; (b) total launch vehicle origin across 316 European payloads [4].

The annual trajectory confirms that foreign launch reliance remained high: 93.0% in 2024, 89.6% in 2025, and 93.5% in 2026 of European payloads were launched on non-European vehicles, even as total European payloads expanded by 44.2%. Bridging this deficit requires rapid hardware qualification of domestic launchers and orbital transfer vehicles. Responsive facilities such as SPARK directly address the testing bottleneck that constrains European launch autonomy.

## 3 Facility Trade-Offs and European Propulsion Diversity

During the initial 18 months of execution, attempting to implement the original top-down engineering model revealed structural obstacles that prompted a reassessment of the facility development strategy.

### 3.1 The Modularity Question

Designing multi-user space propulsion test infrastructure presents an inherent architectural tension between modularity and individual partner requirements. Attempting upfront accommodation for all propellant combinations, pressure regimes, and flow rates creates architectural over-complexity and speculative capital commitments. Sizing manifolds, valves, and relief circuits for extreme upper bounds increases pneumatic latency, thermal mass during chill-down, and certification costs, while risking underutilised capacity that misaligns with commercial demand. Conversely, catering exclusively to individual partner requirements produces fragmented, single-tenant test stands with bespoke plumbing, proprietary instrumentation, and non-standard interfaces that lock out subsequent users.

Dynamic capital allocation aligned with confirmed commercial and research milestones was therefore adopted in preference to pre-investing in unvalidated infrastructure.

### 3.2 Survey: European Propulsion Diversity and Testing Regimes

Commercial launcher startups and exploration developers rarely switch engine cycles once committed; selecting an architecture represents a foundational commitment backed by significant investment in tooling, turbomachinery, and chamber geometry. Consequently, facility planning must accommodate technological diversity across the ecosystem.

Operationally, these challenges stem from the wide diversity of propulsion technologies spanning both NewSpace

ventures and traditional European space actors. A survey of active European space propulsion systems ($N = 37$ operational and developmental programmes; $N = 23$ chemical and hybrid programmes within the accessible $\leq 100$ kN ground testing envelope [5]) reveals significant diversity across non-interchangeable propulsion architectures. Cryogenic methalox accounts for approximately 22% of programmes ($LOX/LCH_4$, including Avio's M10 expander, The Exploration Company's Huracán, Delta Orbit, Pangea Aerospace, and Sirius Space Services), prioritising reusability, reduced coking, and high specific impulse [6]. Semi-cryogenic RP-1 represents roughly 22% (LOX/RP-1, including Rocket Factory Augsburg's Helix, PLD Space's TEPREL family, Skyrora, and Latitude), prioritising high volumetric energy density, packaging efficiency, and simplified ground logistics.

The remaining landscape comprises green storable bipropellants and in-space thrusters (22%, including MaiaSpace's Colibri, The Exploration Company's Mistral, RFA's Fenix, and Dawn Aerospace), legacy storable hypergolics (17%, such as ArianeGroup's BERTA, Avio's RD-843, and Nammo's LEROS and RELIANCE), solid-liquid paraffin and polymer hybrids (13%, including HyImpulse and HyPrSpace's Terminator), and light hydrocarbons (∼4%, such as LOX/Propane for Isar Aerospace) offering clean-burning, storable cryogenic performance with self-pressurisation [5].

While foundational ground testing regimes, such as cryogenic line chill-down, high-pressure gaseous purges, and sea-level thrust chamber firings up to 60 kN, are universally shared across liquid rocket architectures, the underlying propellant choices require fundamentally distinct fluid handling, pressure envelopes, safety zoning, and disposal systems. Rather than building standalone test cells for each vehicle, SPARK maps European propulsion demand directly into modular facility subsections, co-developing core modules with commercial partners while preserving standardised expansion interfaces, as synthesised in Figure 2.

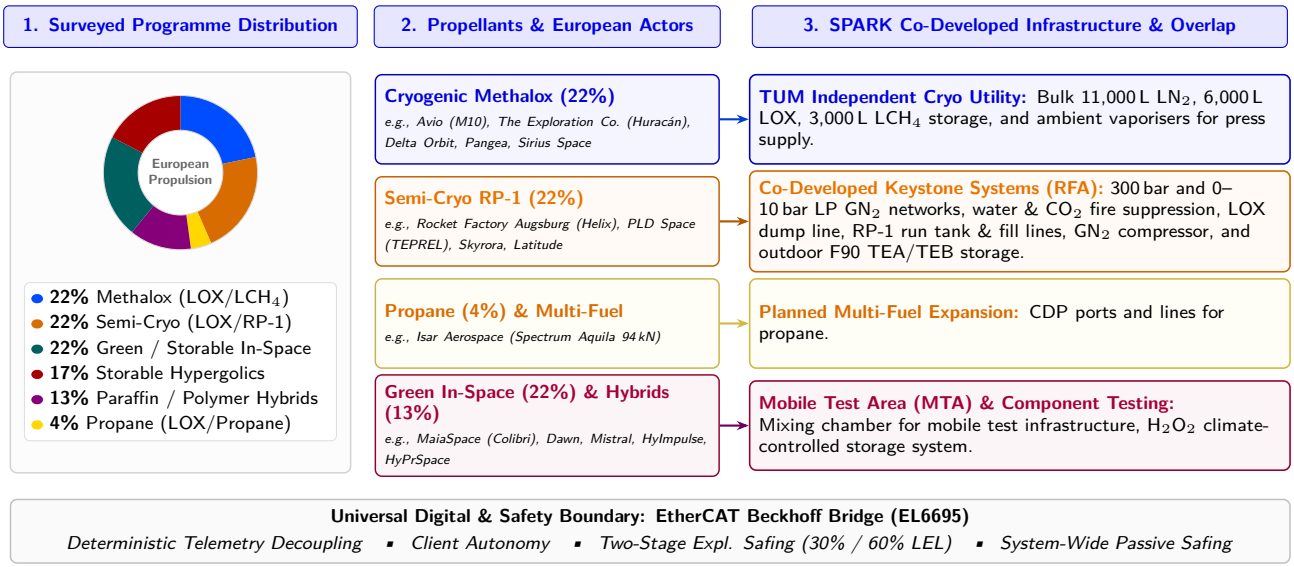



Figure 2: Distribution of propulsion technologies across the European space transportation ecosystem (left) and corresponding modular facility infrastructure co-developed or provisioned under the SPARK Overlap Strategy (right).

## 4 Strategic Adaptation

### 4.1 Modularity Through Co-Development

To reconcile the tension between universal modularity and tailored partner requirements while maximising capital efficiency, SPARK adopted an overlap co-development strategy: partner requirements that align with the long-term facility architecture are implemented as permanent facility assets, with standardised interfaces retained for future expansion. This strategy was implemented through four steps:

1. **Digital Architecture Baseline**: TUM SMP maintains the complete facility architecture modelled within the KiCad P&ID and the 3D CAD facility model, covering target media distribution panels, cryogenic headers, and safety envelopes across all phases.

2. **Requirements Overlap Analysis**: When onboarding an anchor partner, facility engineers cross-reference the client's test article requirements against the long-term roadmap. Overlapping functional blocks (such as bulk cryogenics, high-rate boil-off venting, flow-controlled dump lines, and media feed / storage systems) are identified.

3. **Forward-Compatible Implementation**: Identified overlap subsystems are engineered as permanent facility infrastructure rather than test-specific plumbing, incrementally realising the long-term architecture planned by TUM.

4. **Open Expansion Architecture**: Every co-developed module incorporates standardised mechanical and digital expansion provisions: flanged spool pieces for subsequent propellant lines (e.g., $LCH_4$ or liquid propane headers), blanked relief ports, and unallocated Beckhoff I/O channels. Subsequent users can interface with or expand the existing backbone without modifying prior installations.

### 4.2 Partner Onboarding Process

Partner intake follows a structured four-step onboarding workflow. First, during a Test Scope Definition meeting, the partner defines the test article, operational envelopes (thrust, pressures, temperatures, test durations, etc.), and general compatibility with existing facility infrastructure and capacities is checked.

Second, the user submits a Test Scope, Failure Mode and Effects Analysis (FMEA) document to verify safety compliance, and test article descriptions (e.g., P&IDs). Third, in Modular Service Scoping, service requirements are assembled from modular blocks, separating active testing days, setup, and stand-down periods. Finally, SPARK performs digital interface checks, confirms schedule availability, and issues the campaign agreement.

### 4.3 P&ID as the Authoritative Digital Baseline

Discrepancies between fluid schematics, physical piping, and control software variable assignments cause frequent schedule delays and safety hazards. To prevent them, SPARK established a digital engineering workflow centred on an authoritative Piping and Instrumentation Diagram (P&ID) modelled in KiCad.

KiCad was selected for fluid schematic capture due to its hierarchical sheet structure, strict component tagging, and open-source data architecture. The P&ID models all facility media systems down to individual manual valves, pneumatic actuators, relief valves, and instrumentation. Every component receives a standardised alphanumeric designator ensuring traceable component selection, procurement, and maintenance.

This digital baseline coordinates the facility life cycle:

- **Automated Bill-of-Materials (BOM)**: Pressure ratings, temperature limits, and material compatibility data are embedded as component metadata, accelerating procurement. Component specifications are checked automatically against system state, and incongruities are flagged.
- **Control Software Variable Synchronisation**: Variable tags in the Beckhoff TwinCAT process control system directly mirror KiCad designators, eliminating transcription errors during I/O mapping.
- **Automated Signal List Compilation**: Electrical requirements for each sensor and actuator are compiled automatically into cable schedules and terminal assignments across facility control nodes.
- **Collaborative Change Control**: System modifications requested during partner onboarding are formally evaluated against the P&ID and CAD baseline to check compatibility and system safety before any physical hardware is modified on site.

### 4.4 Integration of Partner Systems via CAD Digital Twin

Coordinating physical installation across contractor teams and external partner hardware required strict spatial configuration control. SPARK maintains a master 3D CAD model of the facility to ensure smooth integration of partner systems.

The master CAD model represents the physical test cell, mounting hardpoints, framing channels, clearance volumes, and media connection points. Testing partners are provided with this master model to integrate their test articles, plumbing brackets, and umbilical interfaces prior to mobilisation, enabling early detection of spatial routing conflicts before hardware arrives on site.

### 4.5 Partner Control Interface

A fundamental challenge in multi-user propulsion centres is interfacing facility infrastructure with diverse customer control systems without forcing clients to compromise proprietary flight avionics or control software.

SPARK resolves this boundary using an EtherCAT Beckhoff Bridge based on an EL6695 bridge terminal, illustrated in Figure 3. The bridge provides a deterministic, galvanically isolated boundary between the facility TwinCAT controller and the partner node. Across this interface, the client transmits propellant conditioning setpoints, line chill-down commands, and run-tank targets. In response, the facility TwinCAT system executes local closed-loop PID control to maintain target conditioning and ullage pressures, streaming calibrated pressure and temperature metrics back to the partner. Simultaneously, it coordinates countdown readiness states and bi-directional emergency abort handshakes, ensuring safety oversight while preserving partner control autonomy.

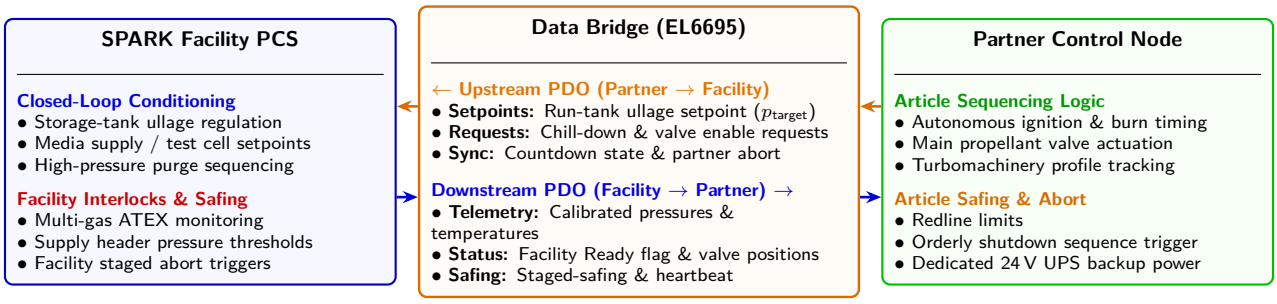



Figure 3: Architecture and cyclic Process Data Object (PDO) exchange of the EtherCAT Bridge interface, providing deterministic telemetry streaming, closed-loop pressure regulation, and synchronised safety abort handshakes between facility and partner nodes.

## 5 Co-Engineering Case Study: Rocket Factory Augsburg

The co-development of infrastructure with Rocket Factory Augsburg demonstrates the SPARK collaborative strategy in practice. Preparing the facility for RFA's bi-propellant liquid oxygen (LOX) and kerosene (RP-1) testing required coordinated fluid, control, and operational adaptations:

### 5.1 Fluid Architecture Co-Engineering

- **Dual-Pressure $GN_2$ Distribution Systems**: High-pressure (300 bar) and low-pressure (0–10 bar) $GN_2$ distribution networks were co-developed. Fed from ambient vaporisers connected to the chair's bulk $LN_2$ storage tank and supplemented by refillable bundles, this integrated system delivers high-pressure propellant tank pressurisation, manifold purges, and component cooling, alongside regulated 0–10 bar pneumatic actuation gas throughout the facility halls. The high-pressure storage can be operated as two separate 6-bundle manifolds in parallel or cross-connected into a unified 12-bundle high-capacity configuration. Additionally, the high-pressure bundles serve as a redundant pneumatic reservoir for actuation and control gas.

- **Water and $CO_2$ Fire Suppression**: Co-developed with RFA, dedicated water and gaseous $CO_2$ fire suppression networks were engineered and routed into the test cell and surrounding hazardous zones to provide a staged fire suppression response that mitigates damage to sensitive electronics and instrumentation.

- **Cryogenic Dump Lines and Common Vents**: To guarantee rapid test stand safing, a dedicated LOX dump line incorporating flow-restricting orifices was installed alongside common facility oxidiser and fuel vent headers, ensuring passive depressurisation of both gas and liquid propellant and pressurant systems.

- **Ignition System Integration (TEA/TEB)**: For engine ignition, a pyrophoric TEA/TEB (triethylaluminium/triethylborane) system was integrated. The pyrophoric medium is housed in an outdoor, F90 fire-rated safety cabinet. From this cabinet, fixed stainless-steel lines are routed directly into the MTA to interface with test articles.

- **RP-1 Storage, Feed Systems, and Run Tank**: In addition to LOX infrastructure, dedicated kerosene storage and feed systems were established, incorporating external supply, feed routing through the main hall, and an integrated 1,000 L RP-1 run tank to support high-flow propulsion testing up to 60 kN.

### 5.2 Synchronised and Staged Article Safing

Integrating industrial test hardware exposed an operational trade-off: original facility safety logic dictates an instantaneous electrical trip upon combustible gas detection, whereas complex test article engines frequently require an orderly shutdown sequence to prevent thermal shock, seal damage, or structural failure, which increases risk exposure during an uncontrolled shutdown.

- **Two-Stage Emergency Shutdown Logic**: TUM and RFA implemented a staged emergency sequence upon sensor limit breach. At the pre-alarm threshold (30% LEL or $O_2 < 19\%$ or $> 23\%$), an automated safing command across the EtherCAT bridge triggers an orderly client-controlled shutdown, isolating propellant valves and purging manifolds. If gas concentrations reach the main alarm threshold (60% LEL or $O_2 < 17\%$ or $> 25\%$), facility power to non-Ex equipment trips immediately for a hard emergency stop.

- **Controller Uninterruptible Power Supply (UPS)**: To guarantee that the test article completes its safing sequence even if facility mains power trips unexpectedly, a dedicated short-duration UPS was integrated for client control nodes.

### 5.3 Collaborative Integration and Shared Operations

To balance commercial testing against academic research, shared operational protocols and procedural demarcations were established. Joint hazard assessments and unified access controls enable industrial integration and partner testing to proceed safely alongside ongoing university research activities. Furthermore, critical infrastructure components co-developed with anchor partners were integrated as permanent facility assets, accelerating flight-readiness for the user while expanding SPARK's forward-compatible baseline for subsequent commercial and academic campaigns.

## 6 Resultant Timeline, Current Status, and Facility Envelopes

### 6.1 Updated Phased Installation Framework

Rather than the original top-down plan, the facility operates on an agile, revenue-reinvested roadmap where each phase delivers immediate testing capability. Table 1 outlines the revised phased framework and current implementation status.

Table 1: Updated SPARK Phased Installation

| Phase | Infrastructure Scope | Key Capabilities & Systems | Status |
|---|---|---|---|
| **Phase 0:** Site Heritage | Alpha Jet hall (500 $m^2$), workshop, control & server rooms, 30 m exhaust tunnel. | Full renovation of the facility; development of ATEX zoning, explosion, and fire safety concepts, multi-gas detection, acquisition of hydrostatic equipment, set up of the shaker. | **Completed** |
| **Phase 1:** Bulk Liquid Storage & Primary Utilities | Bulk liquid & high-pressure bundle storage | 11,000 L $LN_2$, 6,000 L LOX, 3,000 L $LCH_4$, & 2,700 L $LC_3H_8$ storage tanks; ambient $LN_2$ evaporator; Mobile Test Area operations (Delta Orbit, student rocketry, ASCENT). | **Completed** |
| **Phase 2:** Gas Media Supply, Run Tanks & Control System | High- and low-pressure $GN_2$ networks; run vessels; partner control interface; fire suppression & vents. | Commissioning of LOX/$LN_2$ loading systems; 300 bar HP and 0-10 bar LP $GN_2$ manifolds; common oxidiser and fuel vents; water & $CO_2$ fire suppression; LOX dump line; RP-1 & LOX run tanks; TEA/TEB system; control system; OP4 test bench commissioning. | **Active 2026** |
| **Phase 3:** Fuel & MDP Expansion | Multi-propellant infrastructure to Test Position 1 & 2 (TP1, TP2); methane delivery & flare stack installation. | Further development of MDPs 1-5; installation of $LCH_4$ fill line; commissioning of $LCH_4$ & $LC_3H_8$ system; integration of air systems; hydrogen infrastructure expansion. | **Planned** (2026–2027) |

### 6.2 Physical Layout and Commissioned Media Envelopes

The site layout, illustrated in Figure 4, partitions operational areas to maximise multi-user flexibility while maintaining physical and safety separations (Table 2):

Table 2: SPARK Operational Zones, Capabilities, and Test Envelopes

| Operational Area | Primary Function | Operational Scope & Infrastructure |
|---|---|---|
| Test Position 1 (TP1) | Primary Hot-Fire | Aligned axially with exhaust tunnel; rated up to 60 kN. |
| Test Position 2 (TP2) | Subscale Hot-Fire | Subscale engines and in-space thrusters up to 10 kN. |
| Mobile Test Area (MTA) | Mobile Infrastructure | Indoor mixing chamber cell for self-contained partner stands. |
| Vibration Testing Area | Mech. Qualification | Cell housing vibration test rig for launch qualification. |

Media delivery is organised into localised Media Distribution Panels (MDPs) and a central Cryogenic Distribution Panel (CDP). Rigid facility headers supply wall-mounted panels from which users connect via standard tube fittings. Table 3 details the fluid subsystems, storage capacities, and operational envelopes.

### 6.3 Safety Engineering and Airfield Compliance

Because SPARK operates adjacent to the research airport Oberpfaffenhofen, explosion protection and acoustic management formed core design criteria. A formal Explosion Protection Document was prepared in accordance with the German Hazardous Substances Ordinance and ATEX directives. Hazardous zones were classified: Zone 0 (tank interiors and closed vent headers), Zone 1 (immediate release zones at injector faces, disconnects, and relief ports), and Zone 2 (secondary containment areas in the hall and mixing chamber). The RP-1 yard features certified secondary containment bunds and impact-rated bollards.

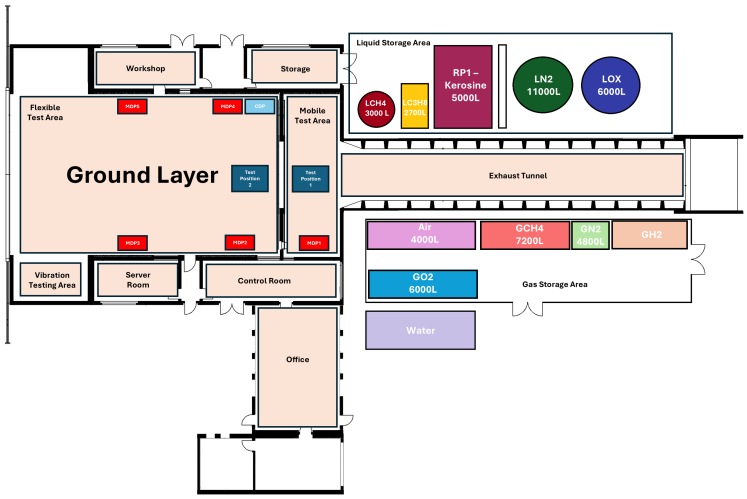



Figure 4: Layout and media supply of the SPARK test facility at the research airport Oberpfaffenhofen, depicting Test Positions 1 and 2, the indoor Mobile Test Area (MTA), 30 m acoustic exhaust tunnel, media distribution panels (MDP1–5), Cryogenic Distribution Panel (CDP), and external bulk propellant storage areas.

Table 3: SPARK Fluid Subsystem Architecture, Storage, and Operational Envelopes

| **Subsystem / Medium** | **Supply / Storage** | **Pressure (bar)** | **Function & Operational Scope** |
|---|---|---|---|
| **In Installation / Operational Systems** | | | |
| Dual-Pressure $GN_2$ | 300 bar Bundles & Evaporator line | 300 / 0-10 | HP tank pressurisation, cooling, purges; LP pneumatic actuation. |
| Bulk & Run LOX | 6,000 L Tank / 2,500 L Tank | 10 | Bulk storage, run tanks, engine propellant feed. |
| Bulk $LN_2$ Utility | 11,000 L Bulk Tank | 10 | System chill-down, cold flow testing, evaporator $GN_2$ feed. |
| RP-1 | Liquid Storage Area / 1,000 L Tank | 10 | Fuel feed into hall run tanks, engine test cells. |
| Ignition (TEA/TEB) | Outdoor Storage | 10 | Hypergolic ignition lines to TP1; nitrogen-purged to roof burn-off barrel (BOB). |
| Vent & Dump Systems | Facility Vents / Flare | Atmospheric | Common GOX/$GN_2$ vents, LOX dump line, hydrocarbon disposal via BOB. |
| **Planned Systems** | | | |
| High-Pressure Air | High-Pressure Buffer | 230 | Run-tank and water-cooling tank pressurisation. |
| Oxygen Gas (GOX) | 300 bar Bundles | 160 | Run-tank pressurisation, gaseous propulsion test feed. |
| Cryogenic Hydrocarbons ($LCH_4$ / $LC_3H_8$) | 3,000 / 2,700 L Tanks | 10 | Run-tanks for cryogenic methalox and liquid propane testing. |

Atmospheric composition is continuously monitored by dedicated methane, propane, and oxygen sensors measuring combustible gas lower explosive limits (LEL) and oxygen concentration. Interfaced with the PCS via fail-safe dry contacts, the gas detection system executes a two-stage emergency safety response. The pre-alarm threshold commands automated safing across the EtherCAT bridge, while the critical main alarm trips non-Ex facility power, isolating propellant headers and enforcing passive normal-safe states.

For acoustic boundary compliance, an acoustic exhaust monitoring system is being implemented to characterise sound emissions and coordinate firings with airfield operations. Attenuation is provided by the 30 m exhaust tunnel lined with concrete baffles and mineral wool dampeners.

## 7 Operational Test Campaigns

Since the EUCASS 2025 presentation, SPARK has transitioned from civil refurbishment into an active propulsion testing centre. Key infrastructure elements across the site are illustrated in Figure 5.



(a) Main test hall (b) 30 m exhaust tunnel (c) Liquid storage

Figure 5: Physical realisation of infrastructure elements at TUM SPARK: (a) main test hall, (b) 30 m acoustic exhaust tunnel, and (c) outdoor cryogenic bulk storage and high-pressure bottle bundles.

### 7.1 OP4 Thrust Chamber Hot-Fire Campaigns

The chair's propulsion test bench, OP4, was relocated from Garching to Test Position 2 at SPARK, with a successful integration into the mobile test platform system [3]. Following hookup of propulsion media and electrical checkout, the bench achieved full operational status, interfacing with test partners' integrated article control systems:

- **Full-Load Hot-Fire Testing**: OP4 completed multiple successful hot-fire test runs with gaseous methane ($GCH_4$) and gaseous oxygen (GOX) at chamber pressures up to 30 bar.
- **TOSCA Campaign**: OP4 hosted the multi-institutional TOSCA campaign, executing high-frequency combustion stability, thermal wall loading, and diagnostic measurements under representative rocket operating conditions, exploring wear and fatigue mechanisms in additively manufactured regenerative cooling systems [7].
- **$CH_4$/LOX Hot-Fires**: OP4 has been adapted to handle liquid oxygen (LOX), soon enabling hot-fire testing of porous injection mechanisms with chamber pressures up to 80 bar.

### 7.2 Academic and Partner Campaigns

The facility hosts a diverse spectrum of institutional demonstrators and commercial test campaigns. Under the ASCENT project, the Chair of Space Mobility and Propulsion is developing TUM's Rocket Hopper test bench, an EU-funded technology test bed. Testing of the ASCENT propulsion system is conducted directly at SPARK, utilising the Mobile Test Area in the mixing chamber. Igniter, injector, and chamber characterisation, as well as static firings, are in progress, pursuing the development of novel control systems and validation of fault detection and identification algorithms.

In parallel, the Scientific Workgroup for Rocketry and Spaceflight (WARR) student team at TUM completed full-vehicle integrated testing of their sounding rockets. The campaign validated an electric-pump-fed bi-propellant LOX/ethanol rocket engine and characterised pump performance.

For commercial operations, the startup Delta Orbit conducts regular testing on the Mobile Test Area using a self-contained test bench. Rather than integrating into the permanent facility media headers, Delta Orbit serves as an operational process partner, collaborating with SPARK to validate agile test procedures, mobile ground logistics, and joint safety protocols for commercial hot-fire operations.

Additionally, the chair's Water Electrolysis Propulsion (WEP) test benches (WEPPY and WALLY) evaluate in-space water-electrolysis micro-thrusters supporting the ESA-funded Ice2Thrust lunar ISRU research programme [8].

### 7.3 Component and Environmental Qualification

Beyond hot-fire operations, SPARK provides specialised mechanical and fluid test services. The facility vibration test rig qualifies flight hardware, engine brackets, and avionics packages under representative random vibration and sinusoidal launch load profiles.

## 8 Transferable Lessons Learned

The co-development and multi-user operation of SPARK provide several key lessons for ground-based propulsion facility engineering:

1. **Living P&ID Over Static Documentation**: Rigid Interface Control Documents become obsolete during rapid iterative facility development required to serve a diverse and agile market. Maintaining an authoritative P&ID baseline in KiCad with automated BOM, cable schedules, and software tag generation reduced engineering effort and maintained consistency between fluid, electrical, and control-system definitions.

2. **Deterministic Digital Bridging**: Forcing partner hardware into facility software sequencers creates technical friction, IP risks, and safety concerns. Implementing the EtherCAT Bridge provided a clean operational boundary: the facility autonomously manages safety and tank conditioning, while the partner retains control of engine firing logic. The exchange ensures smooth integration, operational flexibility, while maintaining a coupled safety response despite the split in control authority.

3. **Reconciling Automatic E-Stop with Article Safing**: Immediate facility-wide electrical trips during emergency stops endanger complex rocket engines requiring orderly shutdown. Implementing two-stage alarm logic (pre-alarm trigger for automated client safing, main alarm for hard E-stop backed by article UPS) ensures personnel safety while allowing orderly shutdown of test hardware.

4. **Demand-Driven Infrastructure Scaling**: Capital allocation is optimised by constructing modular, forward-compatible subsections of the facility directly aligned with active campaign demand. Reinvesting commercial test revenues into phased infrastructure expansion enables sustainable, self-funded scaling to full 60 kN capability.

5. **Decoupling Commercial Tiers from Safety Clearance**: Standardising partner intake through a Test Scope & Clearance Declaration ensures that all users undergo identical safety and hazardous-operations reviews, while commercial tiering remains flexible to capitalisation and test requirements.

## 9 Conclusions

The transition of the SPARK test facility from an engineering concept into an operational, multi-user space propulsion centre advances the European space propulsion testing landscape. By pivoting from a top-down requirements process to an agile collaborative integration methodology, the facility successfully co-developed its infrastructure alongside anchor commercial and academic partners. Utilising a dynamic digital P&ID baseline in KiCad, a distributed TwinCAT process control system, and a deterministic EtherCAT hardware bridge, SPARK offers a repeatable, modular, and cost-effective test environment. With successful hot fires on the OP4 bench, student sounding rocket tests with WARR, commissioning of the ASCENT rocket hopper test bench, ongoing engine development testing with Delta Orbit, and the successful keystone integration with Rocket Factory Augsburg, SPARK demonstrates that university-anchored test infrastructure can decisively accelerate the qualification of next-generation European space propulsion technologies.

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